

"Your Repo Is **Fighting** Your AI Agents"

70% of AI coding tasks fail — not because models are bad, but because **context** is wrong. A few files fix it.

53%

Baseline task success

100%

AI-native repo success

1 afternoon

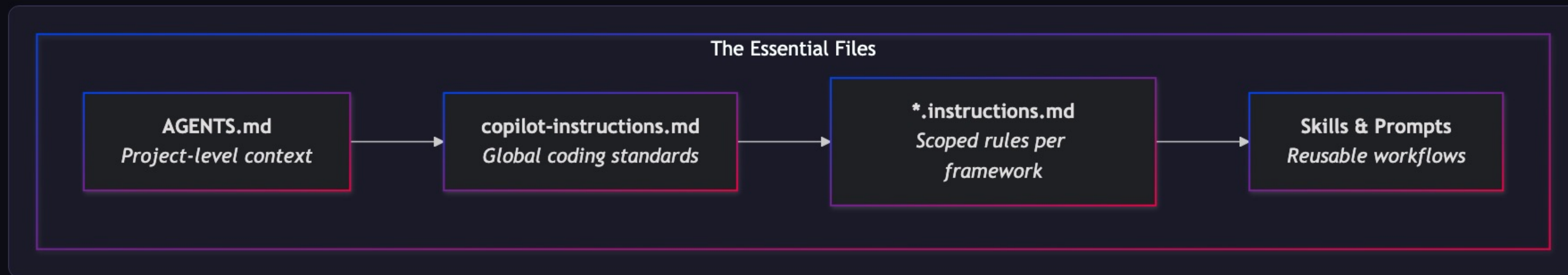
Time to transform



PRACTICAL

The Essential Files — Your AI Agent Toolkit

Four files that every major AI coding agent reads. Add these and your agent stops guessing.

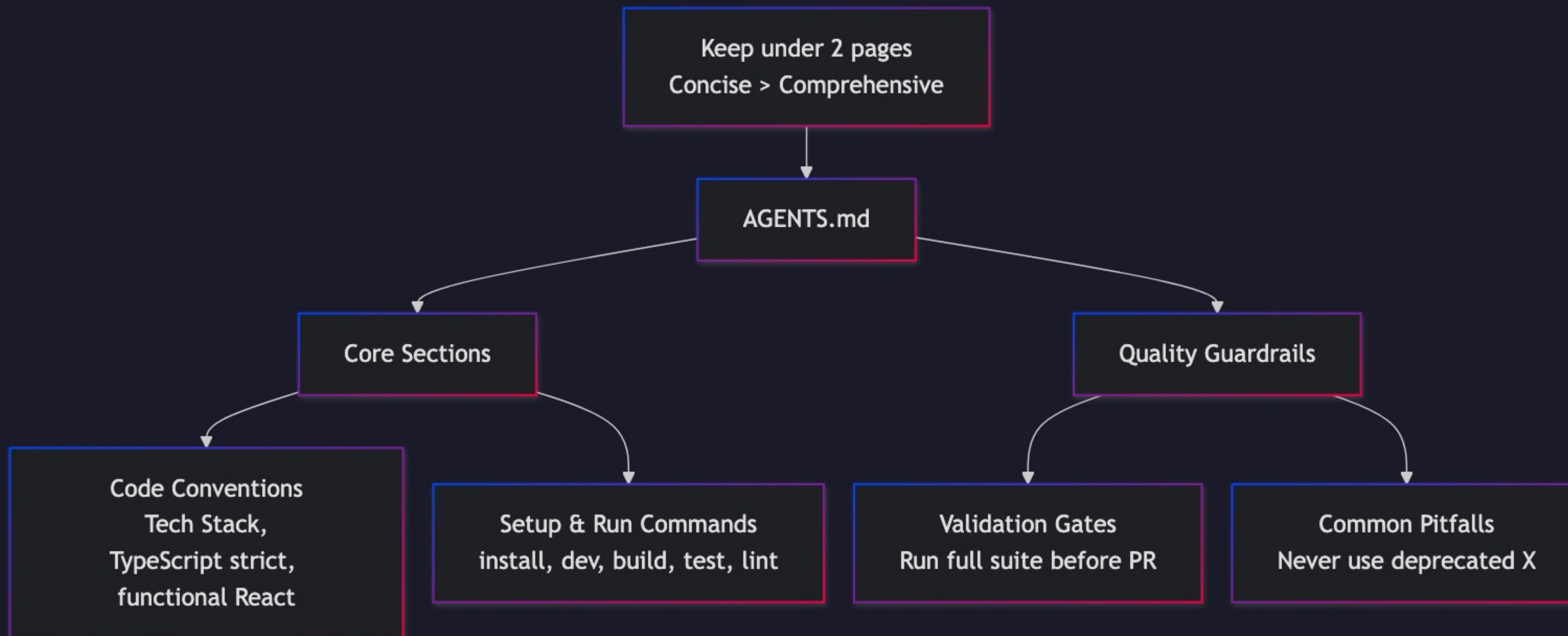


[AGENTS.md Spec](#)

[GitHub Copilot Docs](#)

What Goes in **AGENTS.md**

Five sections. Keep under 2 pages. Concise > Comprehensive. 8KB beat 40KB in Vercel's eval.

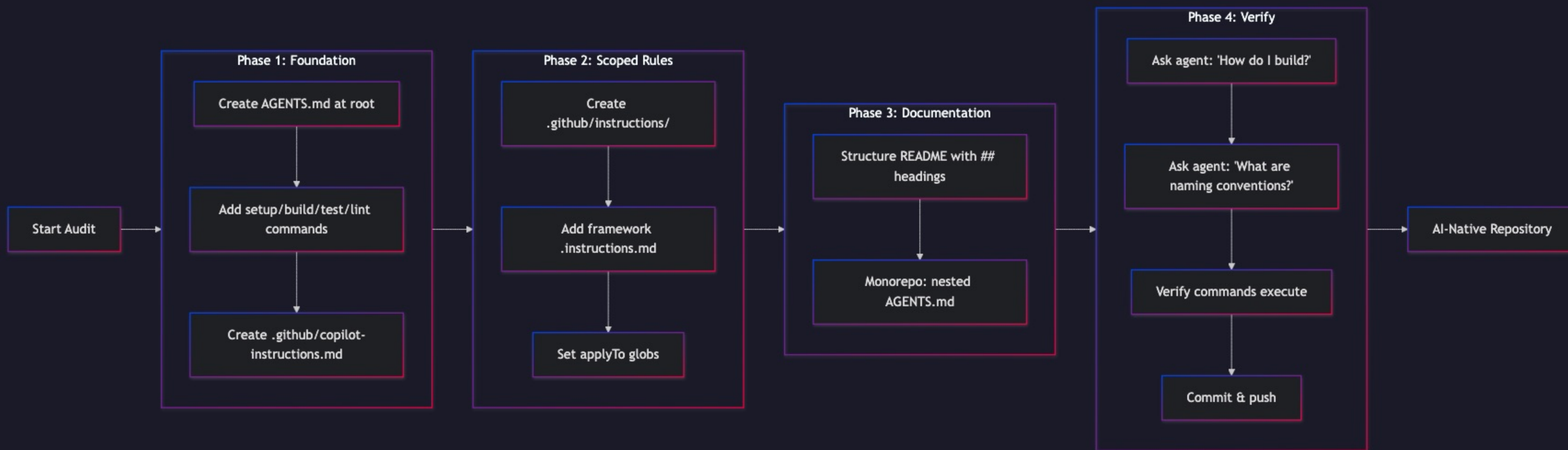


CRITICAL RULE

An 8KB compressed index beat 40KB of full docs. Encode your **tribal knowledge** — the non-obvious conventions, the things that burn people.

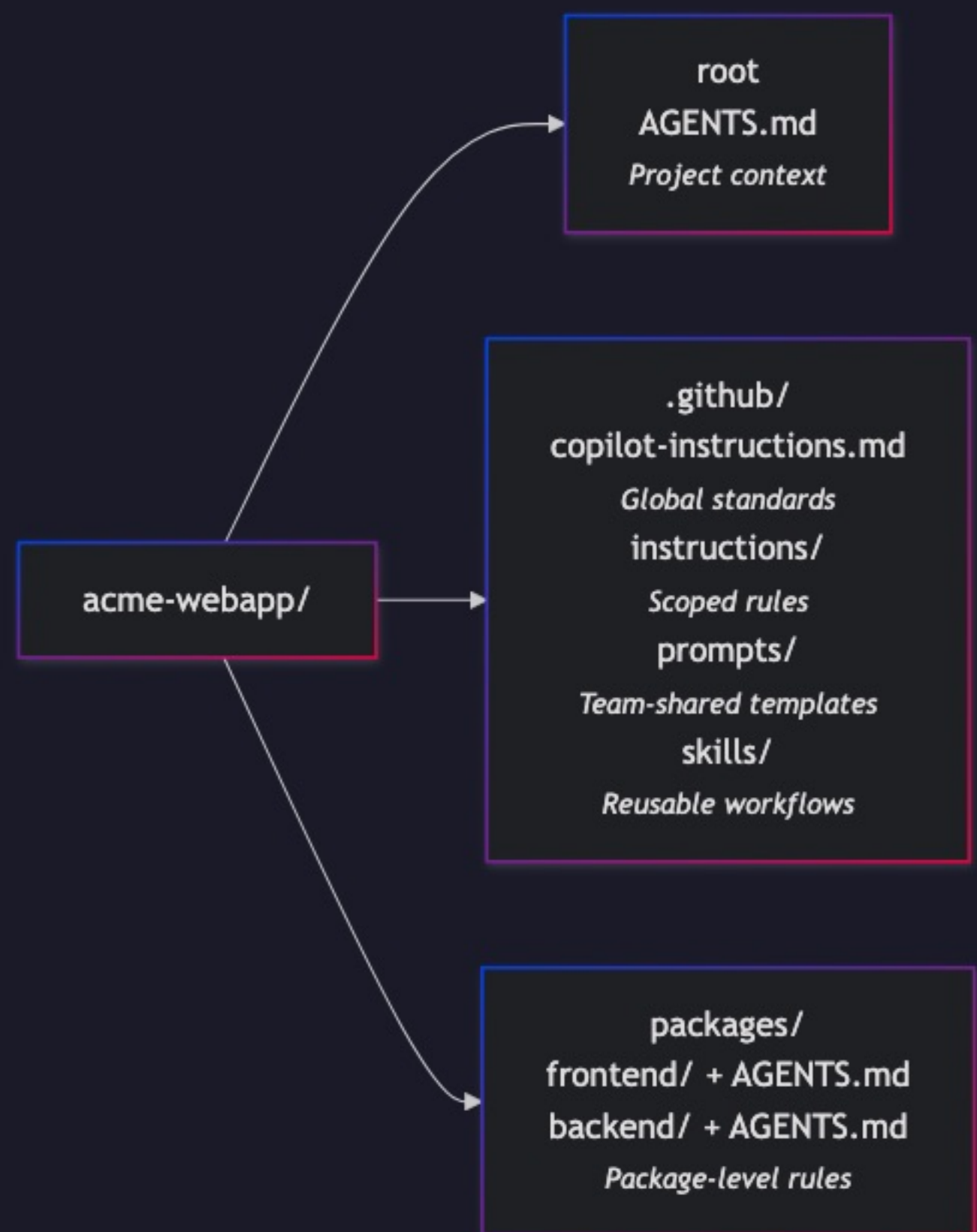
The 15-Minute Audit

Four phases. Foundation → Scoped Rules → Documentation → Verify. You can do this during lunch.



Example Repo in Action

"acme-webapp" — a typical monorepo with all the essential files integrated.

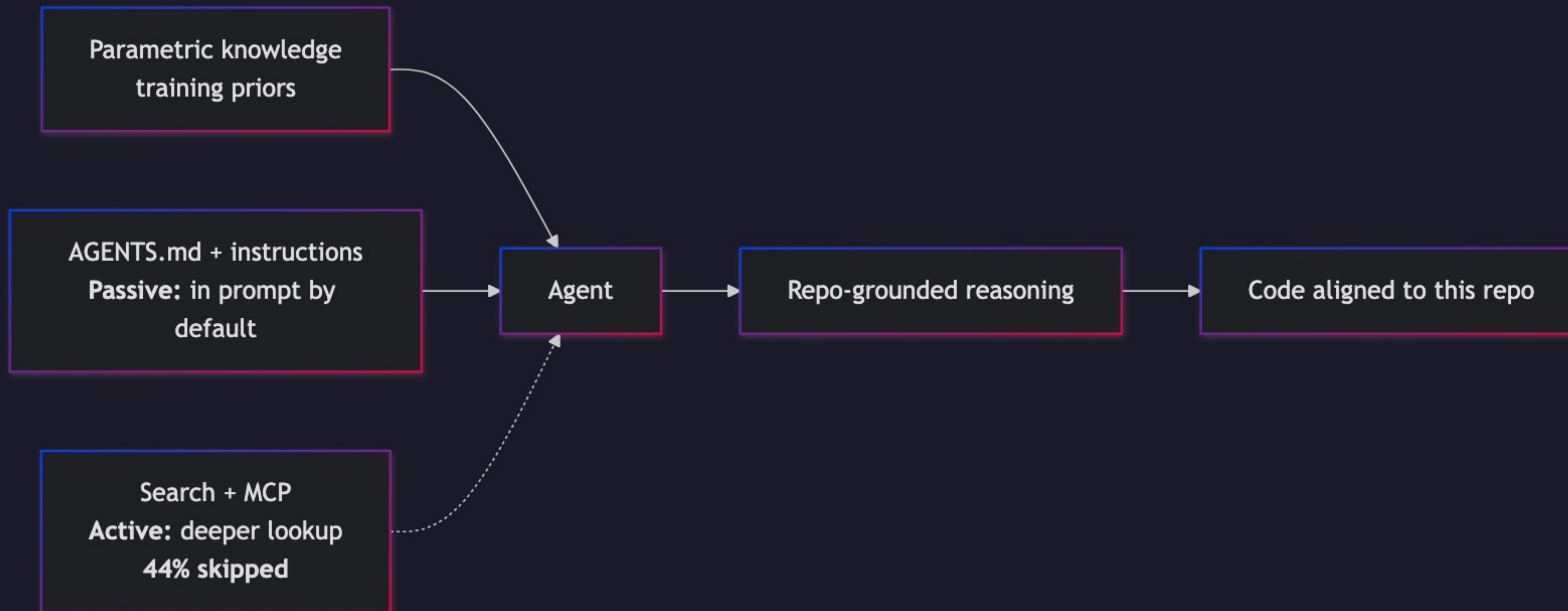


WHAT'S INSIDE

AGENTS.md — project overview & commands. **copilot-instructions.md** — global coding standards. **Scoped rules** — per-framework globs. **Prompts** — team-shared templates. **Skills** — reusable agent workflows.

How Context Changes the Model

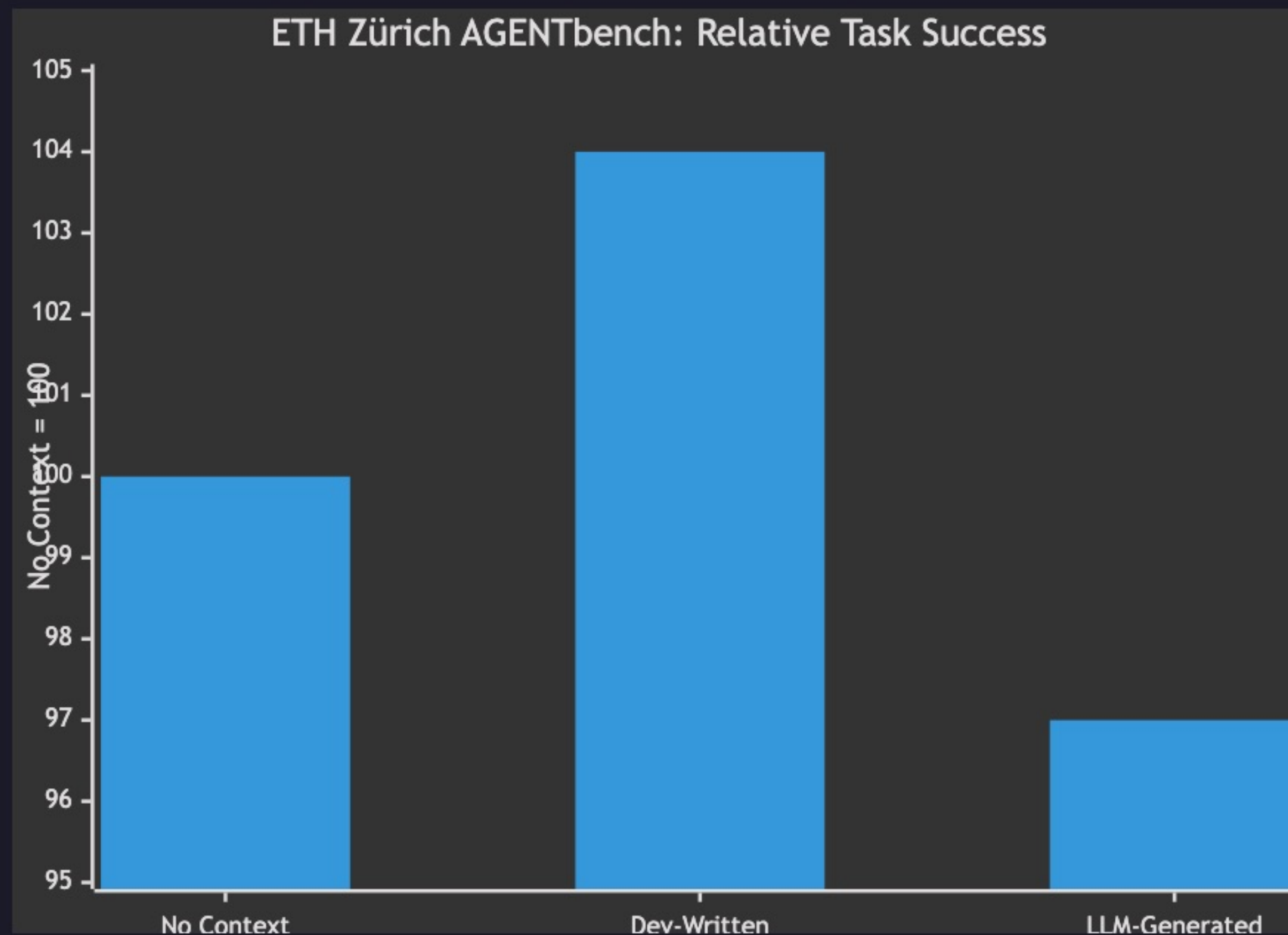
The model is the same. The difference is whether repo rules arrive in the prompt by default or stay hidden behind a retrieval decision.

**KEY INSIGHT**

Passive context gives the agent commands, conventions, and constraints **before** it starts reasoning. Search and MCP still help, but only after the agent decides to use them.

ETH Zürich — What They Actually Found

The strongest public academic study found only modest task-success gains — but clear behavioral changes and higher cost.



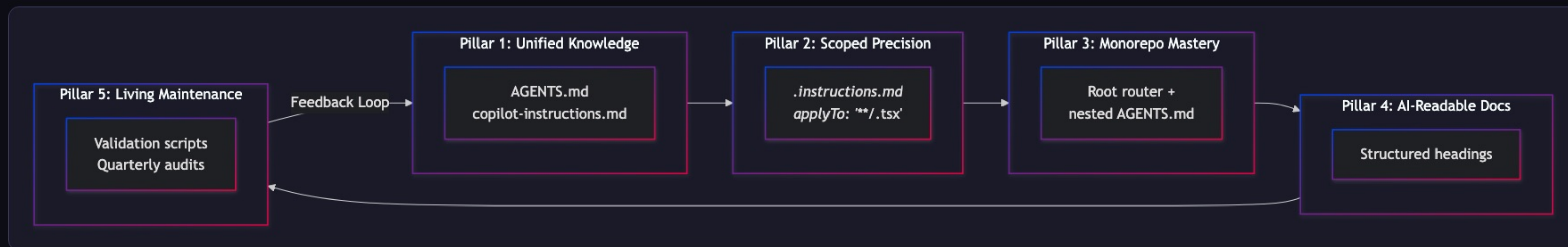
PRACTICAL TAKEAWAY

Minimal, developer-written context looks safer than auto-generated boilerplate. Keep the file lean and focused on non-obvious repo rules.

[arXiv 2602.11988](#)

The Five Pillars of AI-Native Repos

Unified Knowledge → Scoped Precision → Monorepo Mastery → AI-Readable Docs → Living Maintenance



WHERE YOU ARE

You already know how to do Pillars 1–3 from the audit slide. Pillars 4 & 5 are your this-week and this-month work.

Maturity Model & Call to Action

Where is your repo today? Most teams are at Level 1. Getting to Level 3 takes an afternoon.



REFERENCES

Key Citations & Sources

Primary research, specifications, and documentation referenced throughout this talk.

[1] Vercel Blog — *AGENTS.md outperforms skills in our agent evals* (Jan 27, 2026)

vercel.com/blog/agents-md-outperforms-skills-in-our-agent-evals

[2] arXiv 2602.11988 — *Evaluating AGENTS.md: Are Repository-Level Context Files Helpful for Coding Agents?* (ETH Zürich)

arxiv.org/abs/2602.11988

[3] AGENTS.md Specification — Official specification for repository-level agent instruction files

[agents.md](#)

[4] GitHub Docs — *Adding custom instructions for GitHub Copilot*

docs.github.com/.../adding-custom-instructions-for-github-copilot

[5] Anthropic — *The Complete Guide to Building Skills for Claude*

platform.claude.com/.../agent-skills/best-practices

[6] GitHub Blog — *How to write a great agents.md — Lessons from over 2,500 repositories*

github.blog/.../how-to-write-a-great-agents-md...

Context Rot — The Silent Killer

Even million-token windows have a ticking time bomb. Agents silently auto-compact — your conventions are the first things discarded.

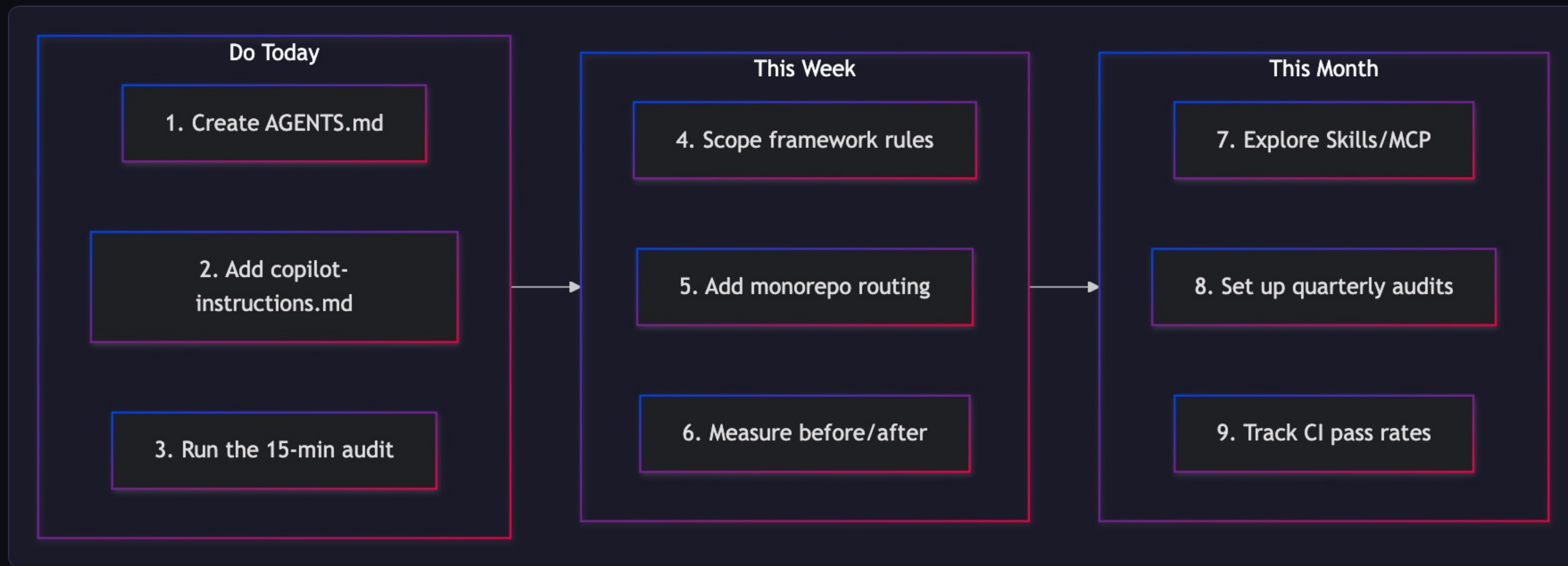


WHY THIS MATTERS

The code compiles. Tests might pass. But it silently violates your architecture. This is why we need **structured context**.

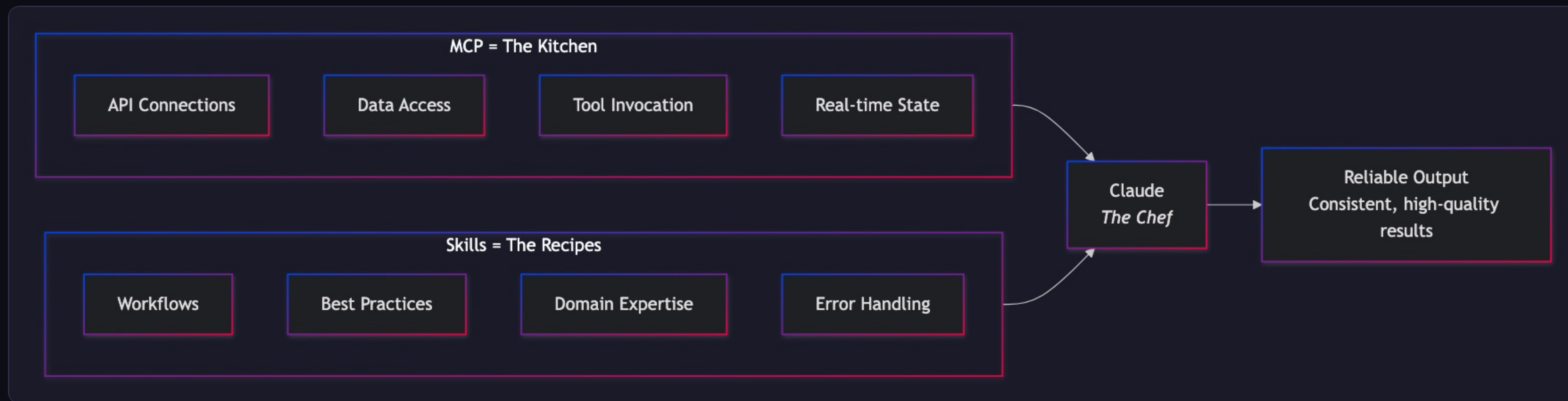
Anthropic's Progressive Disclosure

Three-level loading: YAML frontmatter → SKILL.md body → Linked resources. Hundreds of skills, minimal token overhead.



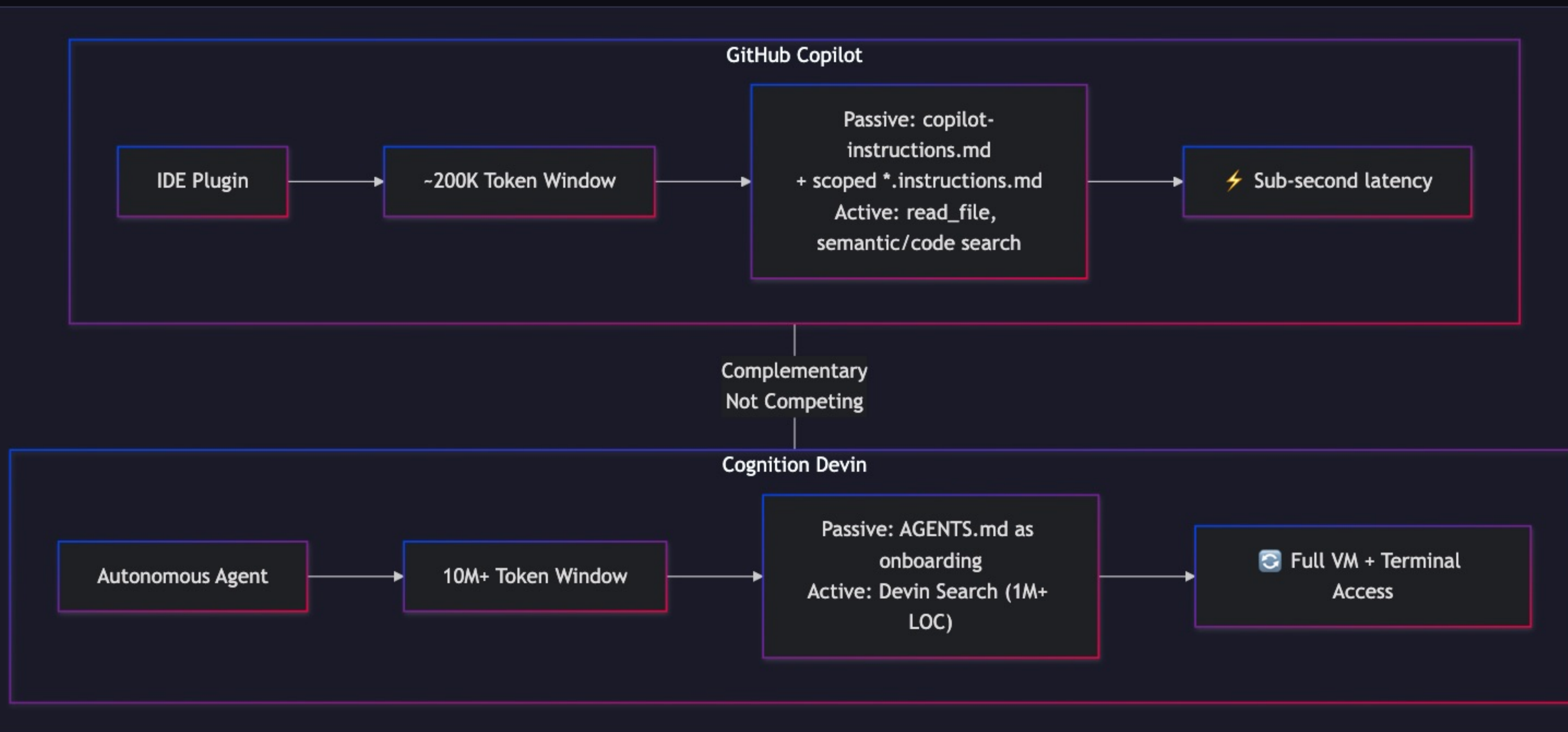
The MCP + Skills Architecture

MCP provides capabilities (the kitchen). Skills provide workflows (the recipes). Together: reliable, scalable AI development.



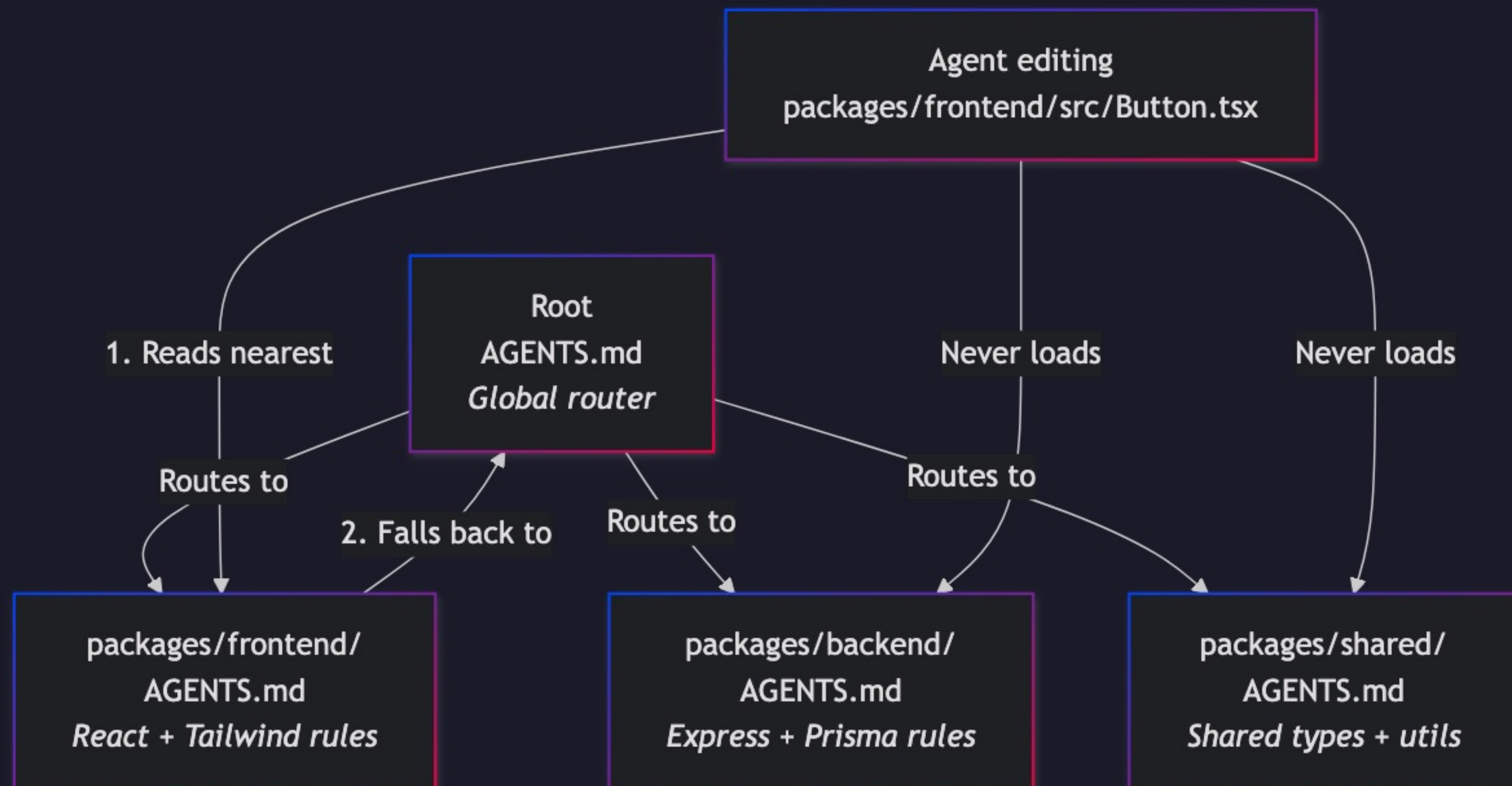
Copilot vs. Devin — Two Architectures

IDE plugin (sub-second, surgical) vs. autonomous cloud agent (full VM, large-scale). Complementary, not competing.



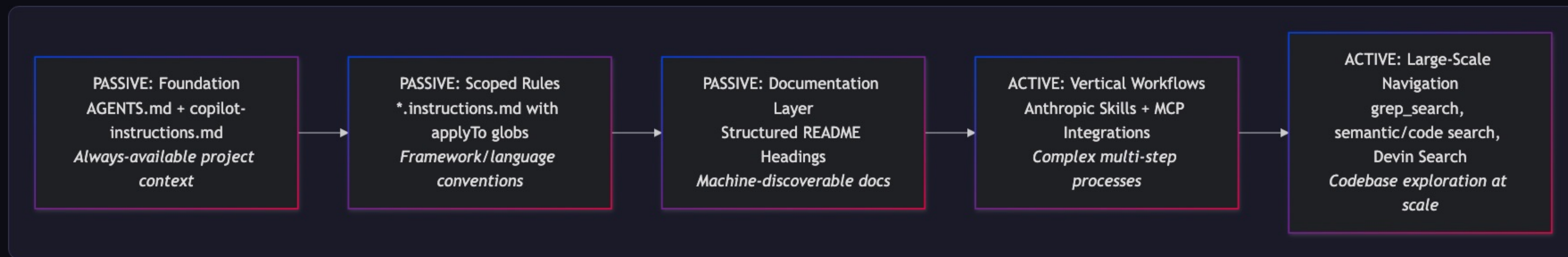
Nearest-File Precedence

The root AGENTS.md routes. Each package overrides. Zero context leakage between domains.



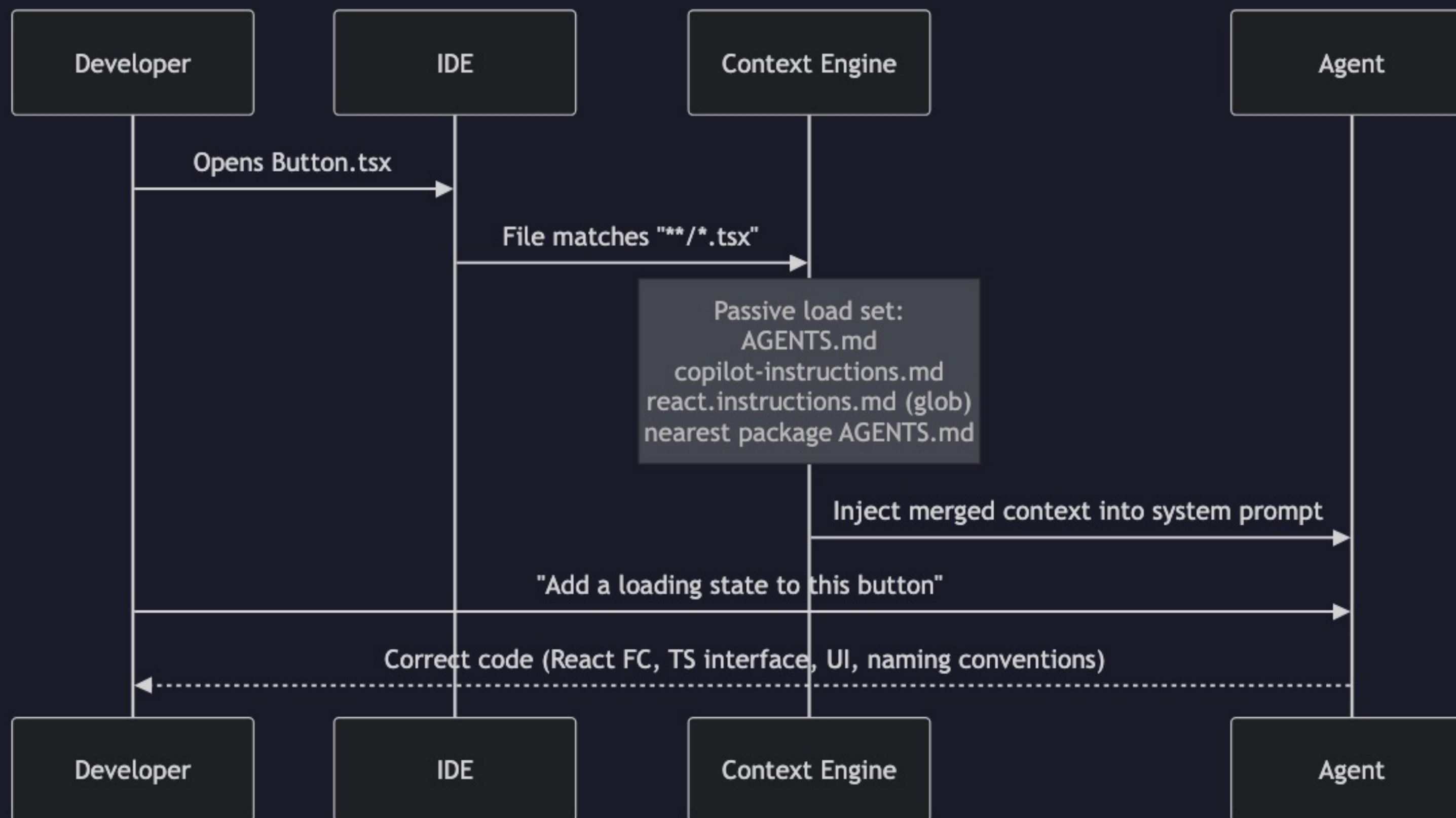
The Hybrid Strategy Stack

Five layers from foundation (passive, always-present) to large-scale active navigation. Build from the bottom up.



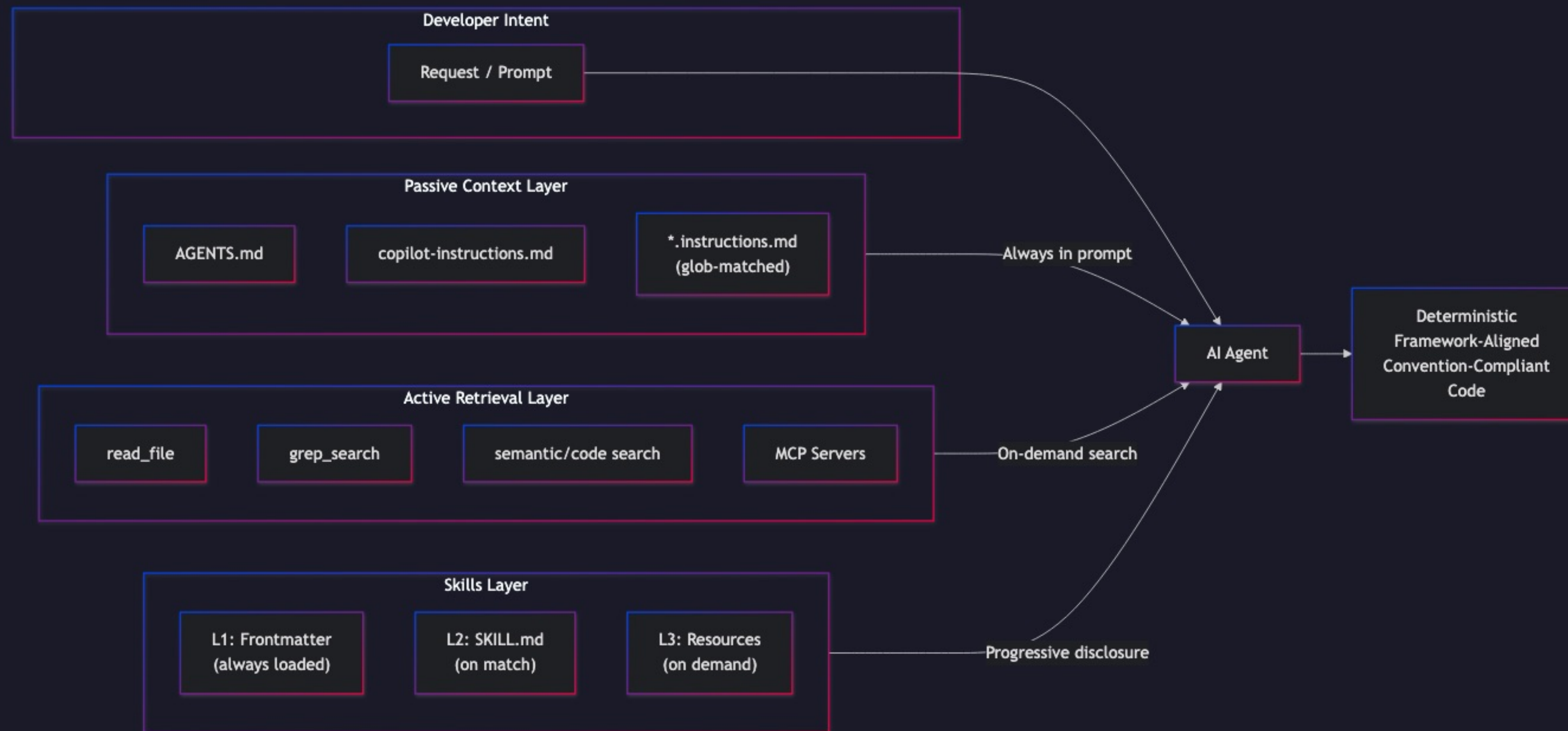
The Passive Steering Workflow

From file open → glob match → context merge → deterministic output.



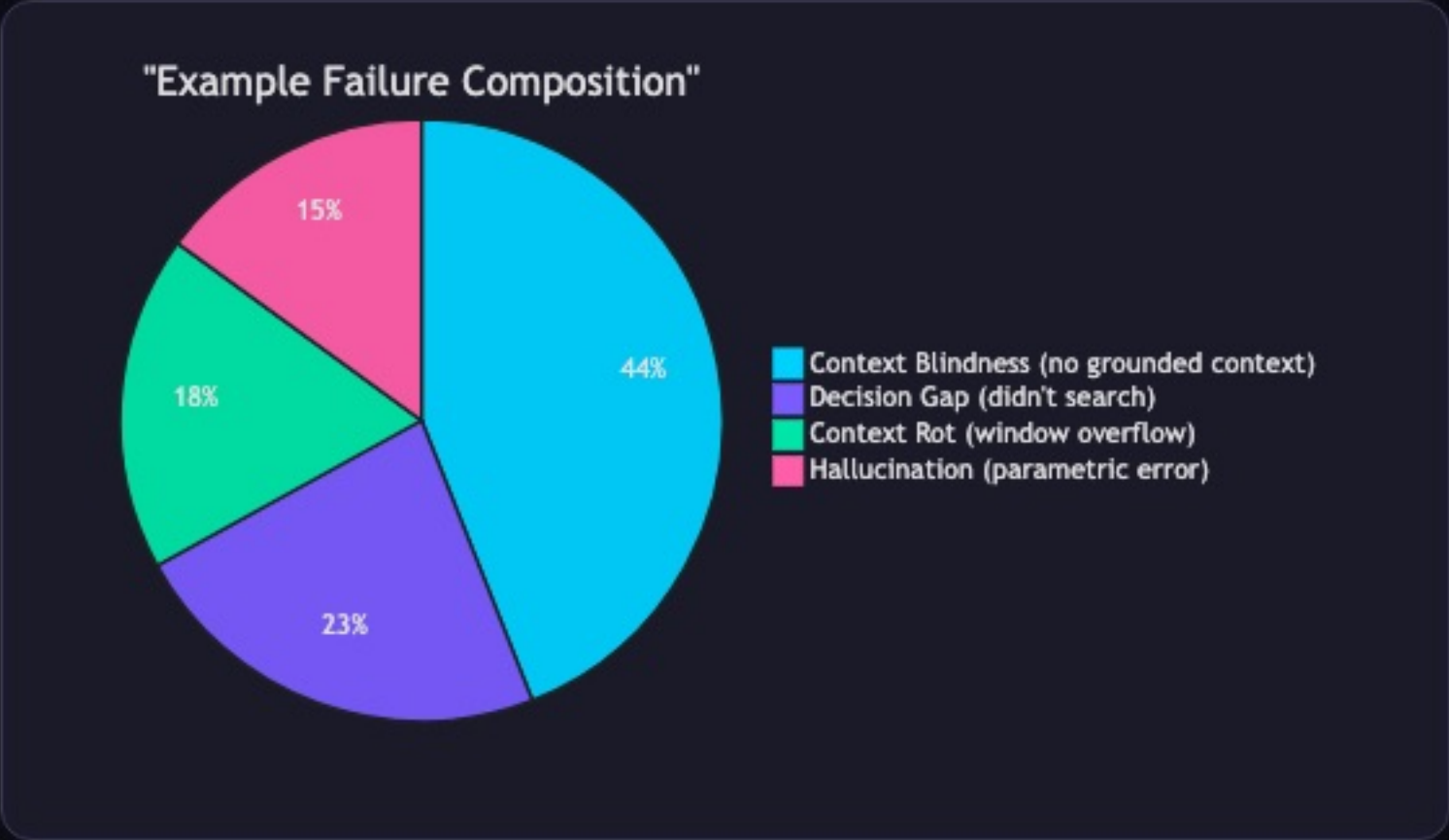
The Complete Context Architecture

Three layers → one agent → deterministic, convention-compliant code.



Key Metrics Summary

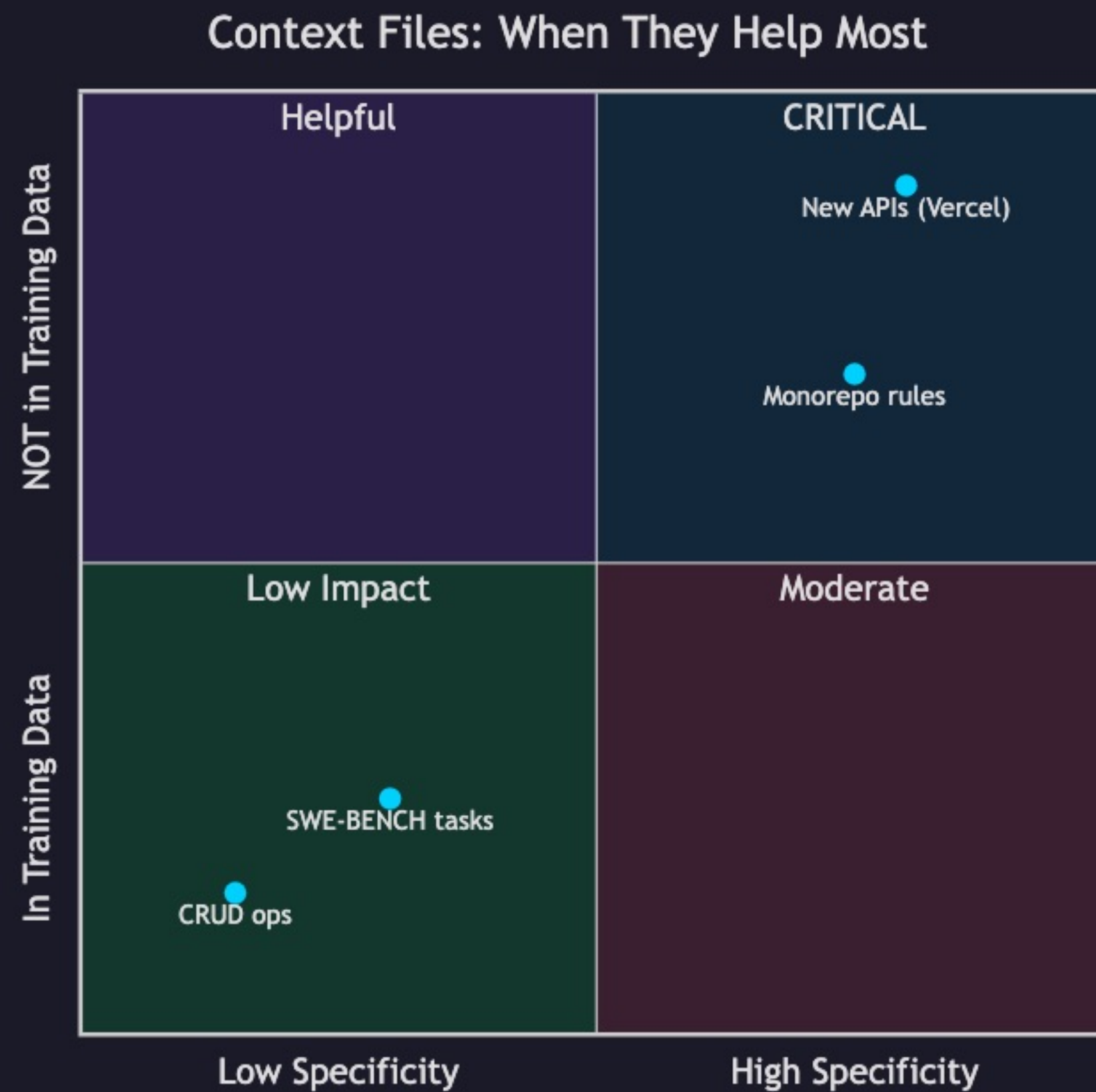
Where agent failures originate — context blindness dominates at 44%.



44%	Context Blindness — no grounded context provided
23%	Decision Gap — agent didn't search for docs
18%	Context Rot — window overflow discards rules
15%	Hallucination — parametric knowledge error

Context File Impact by Scenario

When context files help most: high project specificity + knowledge absent from training data.



INTERPRETATION

Top-right quadrant = **maximum AGENTS.md impact**. Bottom-left = minimal value. This explains the Vercel vs ETH Zürich discrepancy.

Full Presentation Timeline

15 minutes across 3 sections: Practical, The Why, Close — plus Q&A.

